

PORTFOLIO PROJECT SUMMARY

Inventory Management & Ticketing System

for IT Support

Full-Stack Web Application | FastAPI + React + PostgreSQL

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INTERNSHIP ROLE	Backend Development Intern
DURATION	15 Jan 2026 – 29 May 2026 (19 weeks)

100+

REST APIs

5

DB Modules

12

Core Modules

3

User Roles

Abstract

The Inventory Management & Ticketing System (IMS) is a full-stack application built with FastAPI, React, and PostgreSQL to manage IT assets and support tickets on a single centralized platform. It uses a normalized database, role-based access control (RBAC), and secure JWT authentication to ensure data integrity and security. The system streamlines asset tracking, assignment, and audit processes while providing structured ticket management with SLA monitoring and automated alerts. Role-based dashboards deliver real-time insights and performance metrics. With scalable database design and over 100 RESTful APIs, the system improves efficiency, accountability, and decision-making in IT operations.

Confidentiality Note: This project was developed during a professional internship at IT Idol Technologies. As per the company's Non-Disclosure Agreement, the source code, database, and live application are not publicly shareable. This document is a project summary report prepared for portfolio purposes only.

Technology Stack

Layer	Technologies
Frontend	React.js, Tailwind CSS, HTML5
Backend	Python, FastAPI (ASGI, async), Pydantic
Database	PostgreSQL 15+ (3NF normalized, UUID primary keys, soft delete)
DB Migration	Flyway (versioned SQL migrations)
Authentication	JWT + OTP email verification, bcrypt / argon2 password hashing
API Docs & Testing	Swagger (OpenAPI), Postman
Version Control	Git, GitHub

System Architecture

The system follows a modular, layered, scalable architecture across three layers:

- Client Layer** — React.js frontend with role-aware, dynamic rendering of dashboards and forms.
- Backend Layer** — FastAPI handles business logic and REST endpoints: Auth Service, RBAC Middleware, Notification Service, Inventory API, Ticket API, Report API.
- Data Layer** — PostgreSQL 15+, 3NF normalized, UUID primary keys, soft-delete support, immutable audit trails.

Database Design

PostgreSQL 15+ is organized into five functional modules:

Module	Key Tables
Auth	roles, permissions, role_permissions, departments, users
Inventory	asset_categories, asset_sub_categories, inventory_assets, asset_network_details, asset_assignment_history, asset_status_history
Ticketing	sla_policies, tickets, ticket_comments, ticket_status_history
System	notifications

Lifecycle	Flow
Ticket	New → In Progress → Resolved → Closed (Conflict state on escalation)
Asset	Available → Assigned → Under Maintenance → Retired

Core Modules

- Authentication & Access Control
- Dashboard (role-based)
- Inventory Management
- Asset Search
- Employee Search
- Tickets
- Client Assets
- Filters
- Reports & Analytics
- Settings
- My Profile
- User Management

Key Features

- Secure OTP-based signup with JWT-based stateless sessions
- Role-Based Access Control across Admin, Support Engineer, and Employee roles
- Real-time asset and ticket status tracking
- SLA monitoring using timestamp-based ticket lifecycle tracking
- Immutable audit trails for asset assignment and ticket status history
- In-ticket comment threads for requester ↔ support communication
- Auto-generated reports: asset utilization, ticket trends, engineer performance
- Soft-delete mechanism preserving data integrity across all tables

Module Descriptions

Authentication & Access Control — OTP-verified sign-up, JWT-based login, and role assignment (Admin, Support Engineer, Employee) with permission checks enforced at both API and UI level.

Dashboard — Role-based landing view showing asset counts, open/in-progress/closed tickets, conflict status, ticket activity trends, and quick-action shortcuts.

Inventory Management — Add, edit, and track IT assets with category/sub-category classification, purchase details, warranty info, and current assignment status.

Assets — Detailed per-asset view including assignment history, network details (IP, MAC, hostname), and full audit trail of ownership changes.

Tickets — Create, assign, and track support tickets through their full lifecycle (New → In Progress → Resolved → Closed) with priority, type, and file attachments.

Client Assets — Separate tracking for client-owned assets alongside company-owned inventory.

Reports & Analytics — Auto-generated insights on ticket volume, resolution trends, SLA compliance, and asset utilization for data-driven decisions.

User Management — Admin controls for managing user accounts, roles, and department assignments.

Testing Approach

Type	Focus
Functional Testing	Module-wise validation of Auth, Inventory, Tickets, RBAC, and API endpoints via Swagger / Postman
System Testing	End-to-end workflow validation, frontend–backend–database integration, performance, security, cross-brow

Result Comparison

Aspect	Existing System	Proposed System (IMTS)
Asset Tracking	Manual / Spreadsheet	Automated & Centralized
Ticket Management	Email / Verbal	Structured Ticket Lifecycle
Authentication	None / Basic Login	OTP Verified + JWT Secured
Status Visibility	Limited	Real-Time Updates
Access Control	None	Role-Based (Employee / Engineer / Admin)
SLA Tracking	Not Available	Timestamp-Based SLA Monitoring
Communication	Informal	In-Ticket Comment Threads
Reporting	Manual	Auto-Generated Reports & Dashboards

Limitations

- Limited development time restricted implementation of some advanced features
- System is in development phase, not yet fully deployed to production
- Reporting module offers basic analytics with limited data visualization
- Scalability / load testing for large user bases not yet conducted
- Minimal third-party integrations (e.g., email/SMS notification services)

Future Enhancements

- Mobile application development for Android and iOS
- Advanced analytics dashboard with interactive data visualization
- Integration with email / SMS notification services for real-time alerts
- Barcode / QR code integration for faster asset tracking
- Full cloud deployment for improved scalability and reliability

Prepared as a summary of the B.E. Internship Project — Inventory Management & Ticketing System for IT Support, carried out at IT Idol Technologies under the guidance of Mr. Keval Bhagat, in partial fulfilment of the degree of Bachelor of Engineering in Computer Science & Engineering, Gujarat Technological University.